

2012



Mathematics

Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen.
- Write your student number on each booklet.
- Board-approved calculators may be used.
- The mark allocated for each question is listed at the side of the question.
- Marks may be deducted for careless or poorly presented work.

Total Marks – 100

Section I – Pages 2 – 4

10 marks

- Attempt Questions 1 – 10.
- Allow about 15 minutes for this section.
- Answer on the sheet provided.

Section II – Pages 5 – 12

60 marks

- Attempt Questions 11 – 14
- Allow about 2 hours 45 minutes for this section.
- Begin each question in a new booklet.
- Show all necessary working in Questions 11 – 14.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Section I – (10 marks)

Marks

Answer this section on the answer sheet provided at the back of this paper.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

1. The value of $\lim_{x \rightarrow 0} \frac{\sin 4x}{9x}$ is:

- A. $2\frac{1}{4}$
- B. 1
- C. $\frac{4}{9}$
- D. 0

2. For the function $f(x) = 3 \sin^{-1}\left(\frac{x}{4}\right)$ the domain and range of $y = f(x)$ are:

- A. domain $\left\{x: -\frac{3\pi}{2} \leq x \leq \frac{3\pi}{2}, x \in \mathbb{R}\right\}$
range $\{y: -4 \leq y \leq 4, y \in \mathbb{R}\}$
- B. domain $\{x: -1 \leq x \leq 1, x \in \mathbb{R}\}$
range $\{y: -3 \leq y \leq 3, y \in \mathbb{R}\}$
- C. domain $\{x: -3 \leq x \leq 3, x \in \mathbb{R}\}$
range $\left\{y: -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, y \in \mathbb{R}\right\}$
- D. domain $\{x: -4 \leq x \leq 4, x \in \mathbb{R}\}$
range $\left\{y: -\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}, y \in \mathbb{R}\right\}$

3. Solve for x , $\frac{2x+1}{1-x} \geq 1$

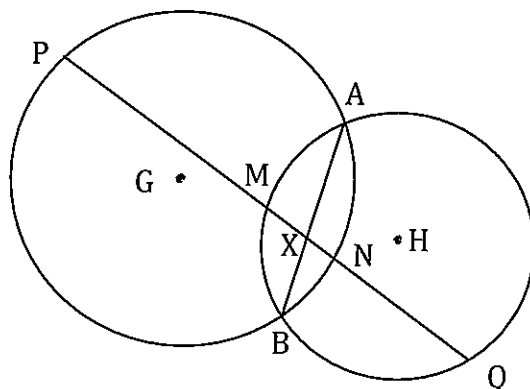
- A. $0 \leq x < 1$
- B. $x \leq 0$ or $x > 1$
- C. $x > 0$ or $x > 1$
- D. $0 < x \leq 1$

Section I (cont'd)

Marks

4. A particle is oscillating in Simple Harmonic Motion where its position x metres from a fixed point O on the same line as its motion after t seconds is given by $x = 2 \cos\left(3t + \frac{\pi}{6}\right)$. What is the maximum speed of the particle?
- A. 2 m/s
B. 6 m/s
C. 0 m/s
D. $\frac{\pi}{9}$ m/s

5.



AB is a common chord to the circles with centres G and H .

PQ is drawn intersecting circle centre G at P and N , intersecting circle centre H at M and Q and intersecting AB at X as shown in the diagram.

If $PM = 18$, $MX = 6$ and $NQ = 15$ then the length NX is:

- A. 5
B. 4
C. 3
D. 2
6. The derivative of $\tan^{-1} \frac{2x}{3}$ is:
- A. $\frac{1}{3+4x^2}$
B. $\frac{1}{\frac{9}{4}+x^2}$
C. $\frac{6}{9+4x^2}$
D. $\frac{3}{4+x^2}$

Section I (cont'd)

Marks

7. The exact value of $\sin^{-1}\left(\cos\frac{2\pi}{3}\right)$ is:
- A. $\frac{\pi}{6}$
B. $-\frac{\pi}{6}$
C. $\frac{\pi}{3}$
D. $-\frac{\pi}{3}$
8. Consider $(1 + 2x)^n$. If the ratio of the coefficient of x^4 to the coefficient of x^6 is 5:8 then the value of n is:
- A. 5
B. 6
C. 7
D. 8
9. The polynomial $P(x) = x^4 - 2x^3 - 7x^2 + 20x - 12$ has a zero of multiplicity 2 at $x =$:
- A. 1
B. -3
C. 2
D. -2
10. A particle moves in a straight line. At time t seconds, where $t \geq 0$, its displacement x metres from the origin and its velocity v metres per second are such that $v = 25 + x^2$.

If $x = 5$ initially, then t is equal to:

- A. $25x + \frac{x^3}{3}$
B. $25x + \frac{x^3}{3} + \frac{500}{3}$
C. $\tan^{-1}\left(\frac{x}{5}\right) - \frac{\pi}{4}$
D. $\frac{1}{5}\tan^{-1}\left(\frac{x}{5}\right) - \frac{\pi}{20}$

Section II – Show all working

Question 11 – Start A New Booklet – (15 marks)

Marks

- a) (i) Find the derivative of $\log_e(\cos^2 x)$

1

(ii) $\int_0^1 \frac{x^2}{x^3 + 1} dx$

1

- b) In the expression of $\left(x^2 + \frac{2}{x}\right)^{10}$ find the coefficient of x^2

1

- c) The quadratic polynomial $ax^2 + bx + 14$ leaves a remainder of -12 when divided by $(x - 1)$, and has $(x + 2)$ as a factor. Find the values of a and b .

2

- d) Find the acute angle between the lines $y = 5 - x$ and $\sqrt{3}y = x + 1$

1

- e) (i) Show that the area of an equilateral triangle of side length x is given by

1

$$A = \frac{\sqrt{3}}{4}x^2$$

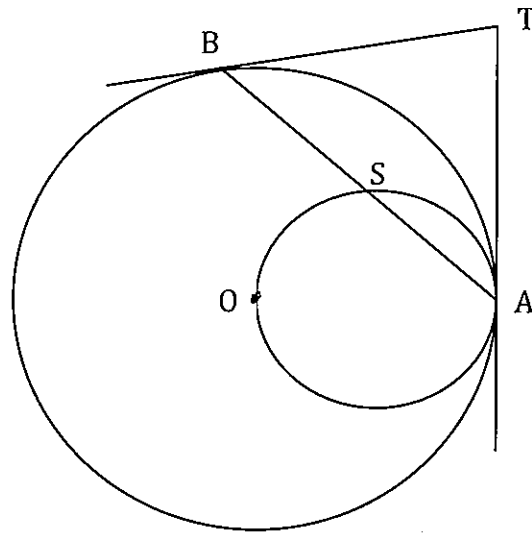
- (ii) The sides of an equilateral triangle are increasing at the rate of 5 mm/s. At what rate is the area of the triangle increasing at the instant the sides are 10 cm long.

2

Question 11 (cont'd)

Marks

f)



Two circles touch internally at a point A and the smaller of the two circles passes through O , the centre of the larger circle.

AB is any chord of the larger circle at S . The tangents to the larger circle at A and B meet at the point T

Prove:

(i) AB is bisected at S .

4

(ii) O, S and T are collinear.

2

Question 12 – Start A New Booklet – (15 marks)

Marks

- a) Use the principle of Mathematical Induction to prove that $7^n + 2$ is divisible by 3 for all positive integers n

2

- b) (i) Show that $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$

1

- (ii) Hence or otherwise find the exact value of:

2

$$\int_0^{\frac{\pi}{6}} \sin 4x \cos 2x \, dx$$

- c) (i) Show that $\frac{d}{dx} (x - \tan^{-1}x) = \frac{x^2}{1+x^2}$

1

- (ii) Hence or otherwise find the exact value of

1

$$\int_0^1 \frac{x^2}{1+x^2} \, dx$$

- d) Given $A(-2, 3)$ and $B(4, 7)$ find the coordinates of the point which divides the interval AB externally in the ratio 3:1

1

- e) If α, β and γ are the roots of $x^3 - 2x^2 + 4x - 7 = 0$ evaluate $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

2

Question 12 (cont'd)

Marks

- f) A particle moving in a straight line has an acceleration given by $\ddot{x} = x^2$ where its displacement is x metres from the origin. If initially the particle is at rest 2 metres from the origin, find its velocity when it is 4 metres from the origin.

2

- g) The normal at $P(2ap, ap^2)$ to the parabola $x^2 = 4ay$ meets the curve again at $Q(2aq, aq^2)$

- (i) Given that the equation of the normal at P is $x + py = ap^3 + 2ap$

1

show that $q = -\frac{(2+p^2)}{p}$

- (ii) Find a value for p so that the lines OP and OQ are at right angles, where O is the origin.

2

Question 13 – Start A New Booklet – (15 marks)

Marks

a) If $3n^2 - 7n + 5 \equiv An(n - 1) + Bn + C$ find A, B and C

2

b) Evaluate, leaving your answer in exact form

3

$$\int_{\frac{1}{8}}^{\frac{\sqrt{3}}{8}} \frac{dx}{\sqrt{1 - 16x^2}}$$

c) If α and β are the roots of $x^2 + bx + c = 0$, form the equation, in general form, whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$

2

d) A curve is defined by the parametric equations $x = t - 3$, $y = t^2 - 9$

(i) Find $\frac{dy}{dx}$ in terms of t

1

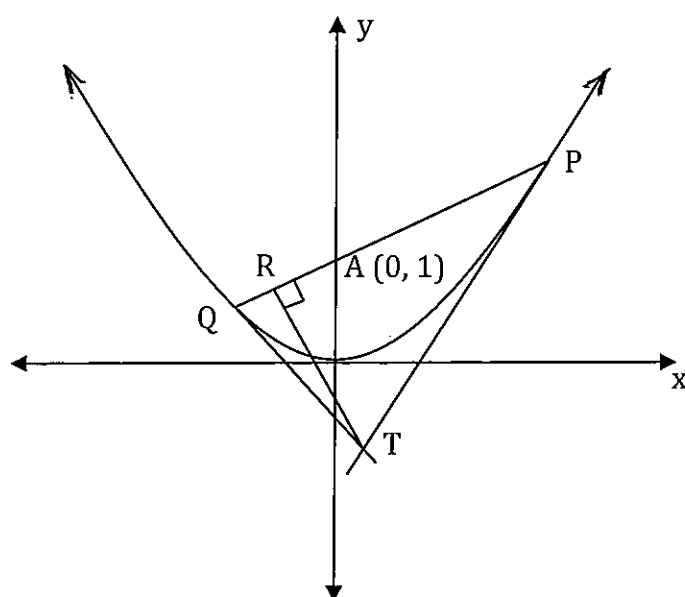
(ii) Find the equation of the tangent to the curve at the point where $t = -3$

2

Question 13 (cont'd)

Marks

e)



PQ is a chord of the parabola $x^2 = 8y$ passing through the point $A(0, 1)$ where P is $(4p, 2p^2)$ and Q is $(4q, 2q^2)$

The tangents to the parabola at P and Q meet at the point T .

R is a point on the chord PQ with $RT \perp PQ$

- (i) Write down the equations of the tangents at P and Q and hence find the coordinates of T

2

- (ii) Show that the equation of the chord PQ is given by

1

$$2y = (p + q)x - 4pq$$

- (iii) Show that $pq = -\frac{1}{2}$

1

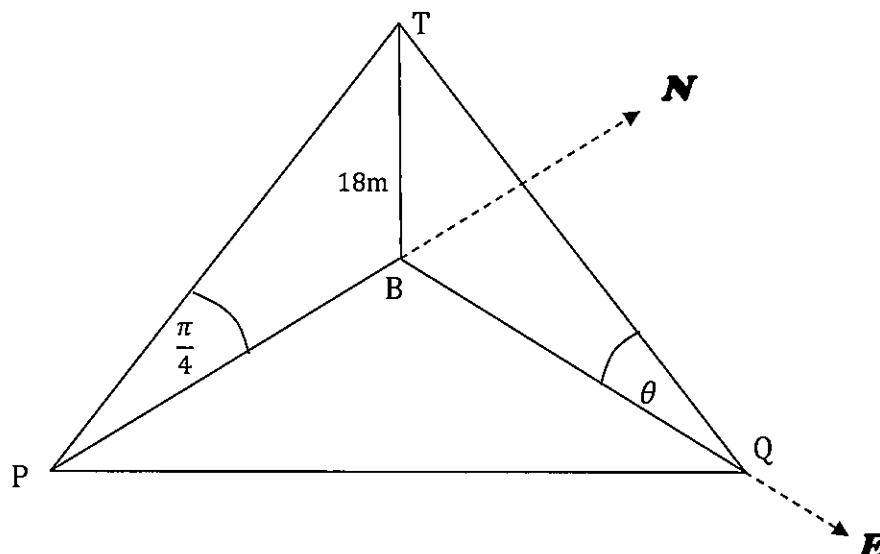
- (iv) Find the equation of RT

1

Question 14 – Start A New Booklet – (15 marks)

Marks

a)



A vertical tower BT of height 18 metres stands with its base B on horizontal grounds. B is due North of a fixed point P and the angle of elevation from P to the top of the tower T is $\frac{\pi}{4}$ radians. Q is a moving point on the ground due East of B and the angle of elevation from Q to T is θ radians where $0 < \theta < \frac{\pi}{2}$. The size of the angle θ is increasing at a constant rate of 0.02 radians per minute.

- (i) Show that $PQ = 18 \operatorname{cosec} \theta$ 2
 - (ii) Find the rate at which the length PQ is changing when $\theta = \frac{\pi}{3}$ 2
- b) A person hits a ball off the ground with a bat, projecting the ball at a velocity of 50 m/s at an angle of projection θ such that $\tan \theta = \frac{3}{4}$
- (i) Taking the origin as the point of projection and $g = 10 \text{ m/s}^2$ show that $\dot{x} = 40$ and $\dot{y} = -10t + 30$ and then find x and y in terms of t 3
 - (ii) A tall building is 100 m from where the ball is hit on horizontal ground. If the ball passes through a small open window in the building find the height of the window above the ground. 2
 - (iii) Find the velocity and angle that the ball makes with the horizontal as it passes through the window. 2

Question 14 continued on next page

Question 14 (cont'd)

Marks

c) Find the general solution in radians of the equation $\sin 2x = \cos x$ 2

d) By considering the expansion of both sides of the identity

$(1 + x)^{m+n} = (1 + x)^m(1 + x)^n$, where m and n are positive integers, show that

$$\binom{m+n}{3} = \binom{m}{3} + \binom{m}{2}\binom{n}{1} + \binom{m}{1}\binom{n}{2} + \binom{n}{3}$$

Student Number: _____ Teacher: _____

Year 12 Mathematics Extension 1 Trial HSC Examination 2012

Section I

Multiple-choice Answer Sheet – Questions 1 – 10

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
correct

-
- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☐ |
| 8. | A ☐ | B ☐ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☐ | D ☐ |

Question 11

$$(a) (i) \frac{d \log_e(\cos^2 x)}{dx} = \frac{2 \cos x (-\sin x)}{\cos^2 x} \quad (\cos x \neq 0)$$

$$= -2 \tan x$$

$$(ii) \int_0^1 \frac{x^2}{x^3+1} dx = \frac{1}{3} \int_0^1 \frac{3x^2}{x^3+1} dx$$

$$= \frac{1}{3} [\log_e(x^3+1)]_0^1$$

$$= \frac{1}{3} (\log_e 2 - \log_e 1)$$

$$= \frac{1}{3} \log_e 2$$

$$(b) T_{k+1} = {}^{10}C_k (x^2)^k \left(\frac{2}{x}\right)^{10-k}$$

$$= {}^{10}C_k x^{2k} 2^{10-k} x^{k-10}$$

$$= {}^{10}C_k 2^{10-k} x^{3k-10}$$

$$\text{If } 3k-10=2$$

$$k=4$$

$$\text{Coeff of } x^2 = {}^{10}C_4 \times 2^6$$

$$(c) \text{ Let } P(x) = ax^2 + bx + 14 \quad \textcircled{1} + \textcircled{2} \quad 3a = -33$$

$$P(1) = -12$$

$$a = -11$$

$$a + b + 14 = -12$$

Subst in ①

$$a + b = -26 \quad \textcircled{1}$$

$$-11 + b = -26$$

$$P(-2) = 0$$

$$b = -15$$

$$4a - 2b + 14 = 0$$

$$4a - 2b = -14$$

$$a = -11 \quad b = -15$$

$$2a - b = -7 \quad \textcircled{2}$$

(d) $y = 5 - x$ $\sqrt{3}y = x + 1$
 $m_1 = -1$ $m_2 = \frac{1}{\sqrt{3}}$

Let θ be the acute angle between the lines

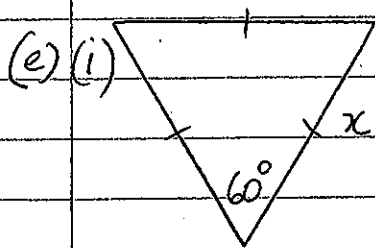
$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-1 - \frac{1}{\sqrt{3}}}{1 + -1 \times \frac{1}{\sqrt{3}}} \right|$$

$$= \left| \frac{-\sqrt{3} - 1}{\sqrt{3} - 1} \right|$$

$$= \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

$$\therefore \theta = 75^\circ$$



$$\begin{aligned} A &= \frac{1}{2} x \times x \sin 60^\circ \\ &= \frac{1}{2} x^2 \times \frac{\sqrt{3}}{2} \\ &= \frac{\sqrt{3} x^2}{4} \end{aligned}$$

(ii) $\frac{dx}{dt} = 0.5 \text{ cm/s}$ $\frac{dA}{dt} = \frac{dA}{dx} \frac{dx}{dt}$

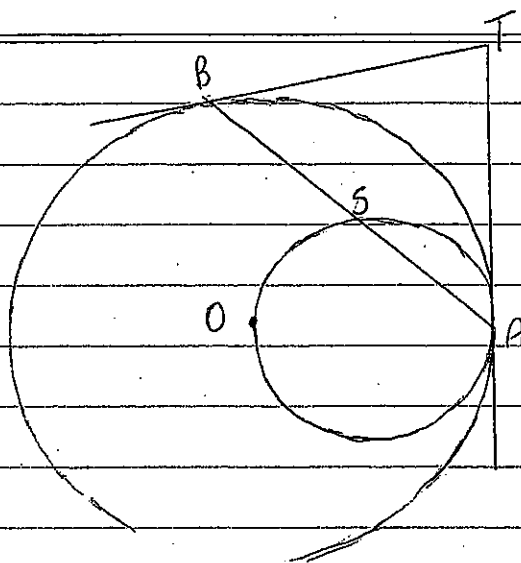
$$= \frac{\sqrt{3} x}{2} \cdot 0.5$$

When $x = 10$ $\frac{dA}{dt} = \frac{\sqrt{3} \times 10}{2} \times 0.5$

$$= \frac{5\sqrt{3}}{2}$$

$\therefore$ Area is increasing at a rate of $\frac{5\sqrt{3}}{2} \text{ cm}^2/\text{s}$ when

$$x = 10$$



Join OA

OA is a radius of larger circle

$\angle OAT = 90^\circ$ (angle between radius and tangent at point of contact)

AT is a tangent to smaller circle

Since $\angle TAO = 90^\circ$ AO must pass through centre of smaller circle

$\therefore$ AO is a diameter of smaller circle.

Join OS $\therefore \angle OSA = 90^\circ$ (angle in a semicircle)

Join OB

OB = OA (radius of larger circle)

$\therefore \triangle OBA$ is isosceles

Since $OS \perp AB$ then OS bisects AB (perp to base of isosceles $\triangle$ bisects base)

ie S is the midpoint of AB

(ii) $\triangle ABT$ is isosceles $BT = AT$ (tangents from external point equal in length)
Since S is midpt of AB then $TS \perp AB$

$$\therefore \angle TSO = 90^\circ + 90^\circ = 180^\circ$$

$\therefore$ TSO is a straight angle

ie O, S, T are collinear

Question 12

(a) Aim : To prove $7^n + 2$ is divisible by 3
ie $7^n + 2 = 3A$ where A is an integer

$$\text{For } n=1 \quad 7^1 + 2 = 9 = 3 \times 3$$

$\therefore$ Proposition true for $n=1$

Assume proposition is true for $n=k$ where k is a positive integer

$$\text{ie } 7^k + 2 = 3B \text{ where } B \text{ is an integer}$$

Aim to show that proposition is then true for $n=k+1$

$$\begin{aligned} 7^{k+1} + 2 &= 7 \times 7^k + 2 \\ &= 7 \times (3B - 2) + 2 \quad (\text{by inductive hypothesis}) \\ &= 7 \times 3B - 14 + 2 \\ &= 3 \times 7B - 12 \\ &= 3(7B - 4) \\ &= 3C \quad (C \text{ is an integer since integers are} \\ &\quad \text{closed under mult}^n \text{ and subtraction}) \end{aligned}$$

ie proposition is true for $n=k+1$ if true for $n=k$.

Hence by induction proposition is true for all positive integers n .

$$\begin{aligned} (b) \text{ (i) } \sin(A+B) + \sin(A-B) &= \sin A \cos B + \cos A \sin B \\ &\quad + \sin A \cos B - \cos A \sin B \\ &= 2 \sin A \cos B \end{aligned}$$

$$\begin{aligned} \text{(ii) } \sin 4x \cos 2x &= \frac{1}{2} (\sin(4x+2x) + \sin(4x-2x)) \\ &= \frac{1}{2} (\sin 6x + \sin 2x) \end{aligned}$$

$$\begin{aligned}
 \therefore \int_0^{\frac{\pi}{6}} \sin 4x \cos 2x \, dx &= \frac{1}{2} \int_0^{\frac{\pi}{6}} \sin 6x + \sin 2x \, dx \\
 &= \frac{1}{2} \left[-\frac{1}{6} \cos 6x - \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}} \\
 &= \left(-\frac{1}{12} \cos \pi - \frac{1}{4} \cos \frac{\pi}{3} \right) - \left(-\frac{1}{12} \cos 0 - \frac{1}{4} \cos 0 \right) \\
 &= \left(-\frac{1}{12} \times -1 - \frac{1}{4} \times \frac{1}{2} \right) - \left(-\frac{1}{12} \times 1 - \frac{1}{4} \times 1 \right) \\
 &= \frac{1}{12} - \frac{1}{8} + \frac{1}{12} + \frac{1}{4} \\
 &= \frac{7}{24}
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad (i) \quad \frac{d}{dx} (x - \tan^{-1} x) &= 1 - \frac{1}{1+x^2} \\
 &= \frac{1+x^2-1}{1+x^2} \\
 &= \frac{x^2}{1+x^2}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \int_0^1 \frac{x^2}{1+x^2} \, dx &= \left[x - \tan^{-1} x \right]_0^1 \\
 &= (1 - \tan^{-1} 1) - (0 - \tan^{-1} 0) \\
 &= 1 - \frac{\pi}{4} - 0 \\
 &= \frac{4-\pi}{4}
 \end{aligned}$$

$$(d) \quad (-2, 3) \quad (4, 7)$$

$$-3 : 1$$

$$\left(\frac{1 \times -2 + -3 \times 4}{-3+1}, \frac{1 \times 3 + -3 \times 7}{-3+1} \right) = \left(\frac{-14}{-2}, \frac{-18}{-2} \right)$$

Point is (7, 9)

(e) $x^3 - 2x^2 + 4x - 7 = 0$ has roots α, β, γ

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \quad \frac{4}{1} = \frac{-7}{1}$$

$$= \frac{4}{7}$$

(f) $\ddot{x} = x^2$

$$\frac{d(\frac{1}{2}v^2)}{dx} = x^2$$

$$\frac{1}{2}v^2 = \frac{x^3}{3} + C_1$$

When $t=0$ $v=0$ $x=2$ OR When $t=0$ $v=0$ $x=-2$

$$0 = \frac{8}{3} + C_1$$

$$0 = -\frac{8}{3} + C_2$$

$$C_1 = -\frac{8}{3}$$

$$C_2 = \frac{8}{3}$$

$$\frac{1}{2}v^2 = \frac{x^3}{3} - \frac{8}{3}$$

$$\frac{1}{2}v^2 = \frac{x^3}{3} + \frac{8}{3}$$

$$v^2 = \frac{2x^3}{3} - \frac{16}{3}$$

$$v^2 = \frac{2x^3}{3} + \frac{16}{3}$$

When $x=4$

$$v^2 = \frac{2 \times 64}{3} - \frac{16}{3}$$

$$= \frac{112}{3}$$

$$v = \pm \sqrt{\frac{112}{3}}$$

But $v \geq 0$

$$\therefore v = \sqrt{\frac{112}{3}}$$

Since $\ddot{x} \geq 0$ ($\ddot{x} = x^2$)

and $v=0$ when $t=0$
particle will always move
in a positive direction

$$\therefore x \neq -4$$

ie When $x=4$

$$v^2 = \frac{2 \times 64}{3} + \frac{16}{3}$$

$$= \frac{144}{3}$$

$$v = \sqrt{\frac{144}{3}}$$

$$= \sqrt{48} = 4\sqrt{3}$$

(9) (i) $x + py = ap^3 + 2ap$
 meets $x^2 = 4ay$
 $x + p \cdot \frac{x^2}{4a} = ap^3 + 2ap$

$$\frac{p}{4a} x^2 + x - (ap^3 + 2ap) = 0$$

has roots $2ap$ and $2aq$

$$2ap + 2aq = -\frac{1}{p/4a}$$

$$= -\frac{4a}{p}$$

$$2aq = -\frac{4a}{p} - 2ap$$

$$q = -\frac{2}{p} - p$$

$$= -\frac{2+p^2}{p}$$

$$= -\frac{(2+p^2)}{p}$$

(ii) $\text{Grad } OP = \frac{ap^2 - 0}{2ap - 0}$
 $= \frac{p}{2}$

$$\text{Grad } OQ = \frac{q}{2}$$

if $OP \perp OQ$ then $\frac{p}{2} \times \frac{q}{2} = -1$

$$pq = -4$$

$$\therefore q = -\frac{4}{p}$$

$$\therefore \frac{-4}{p} = -\frac{(2+p^2)}{p}$$

$$2+p^2 = 4$$

$$p^2 = 2$$

$$p = \pm\sqrt{2}$$

Question 13

(a) $3n^2 - 7n + 5 \equiv An(n-1) + Bn + C$

2 marks

Let $n=0$ $5 = 0 + 0 + C$ $C=5$

Coeff n^2 $3 = A$ $A=3$

Let $n=1$ $3 - 7 + 5 = 0 + B + C$

$1 = B + 5$

$B = -4$

(b) $\int_{\frac{1}{8}}^{\frac{\sqrt{3}}{8}} \frac{dx}{\sqrt{1-16x^2}} = \int_{\frac{1}{8}}^{\frac{\sqrt{3}}{8}} \frac{dx}{4\sqrt{\frac{1}{16}-x^2}}$ 3 marks

$= \frac{1}{4} \left[\sin^{-1} \frac{x}{\frac{1}{4}} \right]_{\frac{1}{8}}^{\frac{\sqrt{3}}{8}}$

$= \frac{1}{4} \left[\sin^{-1} 4x \right]_{\frac{1}{8}}^{\frac{\sqrt{3}}{8}}$

$= \frac{1}{4} \left(\sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2} \right)$

$= \frac{1}{4} \left(\frac{\pi}{3} - \frac{\pi}{6} \right)$

$= \frac{\pi}{24}$

(c) $x^2 + bx + c = 0$

2 marks

$\alpha + \beta = -b$

equation of quad
with $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$

$\alpha\beta = \frac{c}{1}$

$$\begin{aligned}\frac{\alpha}{\beta} + \frac{\beta}{\alpha} &= \frac{\alpha^2 + \beta^2}{\alpha\beta} \\ &= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} \\ &= \frac{u^2 - 2c}{c}\end{aligned}$$

$$\frac{\alpha}{\beta} - \frac{\beta}{\alpha} = 1$$

$$\text{Eqn is } \left(x - \frac{\alpha}{\beta}\right) \left(x - \frac{\beta}{\alpha}\right) = 0$$

$$x^2 - \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)x + \frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 0$$

$$x^2 - \frac{(u^2 - 2c)}{c}x + 1 = 0$$

$$cx^2 - (u^2 - 2c)x + c = 0$$

(Q) Eqn of tangents at P and Q

$$y = px - 2p^2 \quad (1)$$

$$y = qx - 2q^2 \quad (2)$$

Subst (1) in (2)

$$px - 2p^2 = qx - 2q^2$$

$$(p - q)x = 2(p^2 - q^2)$$

$$\begin{aligned}x &= \frac{2(p - q)(p + q)}{p - q} \quad (p \neq q) \\ &= 2(p + q)\end{aligned}$$

$$\begin{aligned}y &= p \cdot 2(p + q) - 2p^2 \\ &= 2pq\end{aligned}$$

$\therefore T$ is point $(2(p + q), 2pq)$

2 marks.

Write equations of
tangent
wrt P, Q

$$\begin{aligned}
 \text{(ii) Grad PQ} &= \frac{2p^2 - 2q^2}{4p - 4q} \\
 &= \frac{2(p-q)(p+q)}{4(p-q)} \\
 &= \frac{p+q}{2}
 \end{aligned}$$

Eqⁿ of chord PQ is

$$y - 2p^2 = \frac{p+q}{2} (x - 4p)$$

$$= \frac{p+q}{2} \cdot x - 2p(p+q) \quad \text{1 mark}$$

$$= \frac{p+q}{2} x - 2p^2 - 2pq$$

$$y = \frac{p+q}{2} x - 2pq$$

$$2y = (p+q)x - 4pq$$

(iii) Since PQ passes through A(0,1)

$$2 = 0 - 4pq$$

$$pq = -\frac{1}{2}$$

1 mark

(iv) RT $\perp$ PQ

$$\therefore \text{Grad RT} = -\frac{2}{p+q}$$

1 mark

equation RT

$\therefore$ Eqⁿ of RT is

$$y - 2pq = -\frac{2}{p+q} (x - 2(p+q))$$

$$y - 2x - \frac{1}{2} = -\frac{2}{p+q} x + 4$$

$$y = -\frac{2}{p+q} x + 3$$

(i) $x = t - 3$ $y = t^2 - 9$

1 mark

$$\frac{dx}{dt} = 1$$

$$\frac{dy}{dt} = 2t$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$= \frac{2t}{1}$$

(ii) When $t = -3$

2 marks

$$x = -6$$

$$y = 0$$

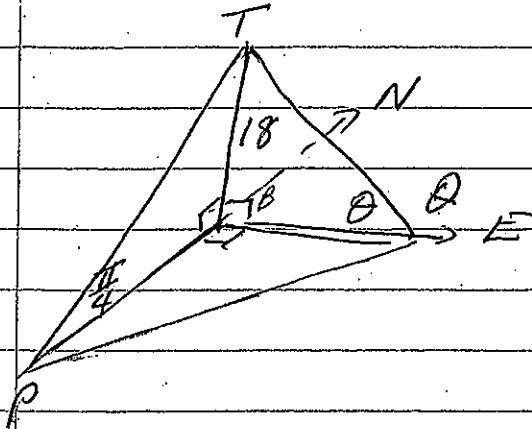
$$\frac{dy}{dx} = -6$$

Eqⁿ of tangent is

$$y - 0 = -6(x - (-6))$$

$$y = -6x - 36$$

Question 14



$$(i) \quad \frac{BQ}{18} = \cot \theta$$

$$BQ = 18 \cot \theta$$

$$\frac{BP}{18} = \tan \frac{\pi}{4}$$

$$= 1$$

$$BP = 18$$

$$\begin{aligned} \text{In } \triangle PBQ \quad PQ^2 &= BP^2 + BQ^2 \\ &= 18^2 + 18^2 \cot^2 \theta \\ &= 18^2 (1 + \cot^2 \theta) \\ &= 18^2 \operatorname{cosec}^2 \theta \end{aligned}$$

$$PQ = 18 \operatorname{cosec} \theta$$

$$(ii) \quad \frac{dPQ}{dt} = \frac{dPQ}{d\theta} \cdot \frac{d\theta}{dt}$$

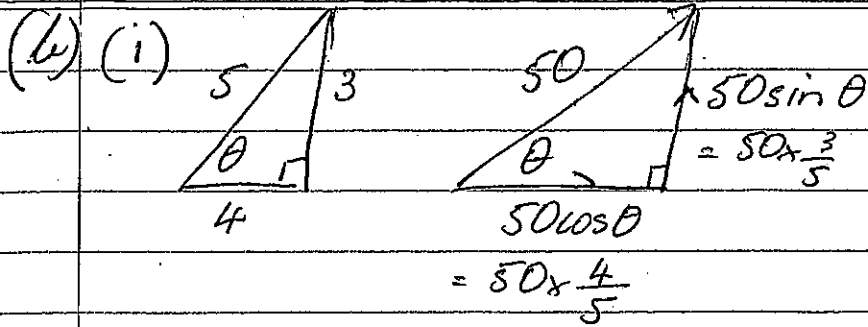
$$= -18 \operatorname{cosec} \theta \cot \theta \cdot \frac{d\theta}{dt}$$

$$= -18 \operatorname{cosec} \theta \cot \theta \cdot 0.02$$

$$= -0.36 \times \operatorname{cosec} \frac{\pi}{3} \cdot \cot \frac{\pi}{3}$$

$$= -0.36 \times \frac{2}{\sqrt{3}} \times \frac{1}{\sqrt{3}}$$

$$= -0.24$$



$$\ddot{x} = 0 \quad \ddot{y} = -10$$

$$\dot{x} = C_1 \quad \dot{y} = -10t + C_2$$

When $t=0$ $\dot{x} = 50 \cos \theta$ When $t=0$ $\dot{y} = 50 \sin \theta$

$$= 40 \quad = 30$$

$$\therefore = C_1 \quad = C_2$$

$$\dot{x} = 40 \quad \dot{y} = -10t + 30$$

$$x = 40t + C_3 \quad y = -5t^2 + 30t + C_4$$

When $t=0$ $x=0$ When $t=0$ $y=0 \therefore$

$$\therefore C_3 = 0 \quad \therefore C_4 = 0$$

$$x = 40t \quad y = -5t^2 + 30t$$

(ii) If $x = 100$ then

$$100 = 40t$$

$$t = \frac{5}{2}$$

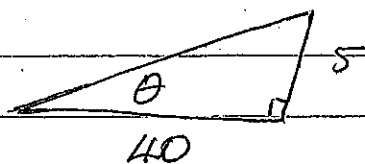
$$y = -5 \times \left(\frac{5}{2}\right)^2 + 30 \times \frac{5}{2}$$

$$= \frac{175}{4}$$

$\therefore$ Height of window is 43.75m

(iii) When $t = \frac{5}{2}$ $\dot{x} = 40$

$$\dot{y} = -10 \times \frac{5}{2} + 30 = 5$$



$$v^2 = 40^2 + 5^2$$

$$= 1600 + 25$$

$$= 1625$$

$$v = \sqrt{1625}$$

$$= 5\sqrt{65}$$

$$\tan \theta = \frac{5}{40}$$

$$= \frac{1}{8}$$

$$\theta = \tan^{-1}\left(\frac{1}{8}\right)$$

$$= 7^\circ 8'$$

Velocity is $5\sqrt{65}$ m/s and angle ball's path makes with the horizontal is $7^\circ 8'$

(c) $\sin 2x = \cos x$

$$2\sin x \cos x - \cos x = 0$$

$$\cos x (2\sin x - 1) = 0$$

$$\cos x = 0 \quad \text{or} \quad \sin x = \frac{1}{2}$$

$$x = \frac{\pm \pi}{2}, \frac{\pm 3\pi}{2}, \frac{\pm 5\pi}{2}, \dots \quad x = \frac{\pi}{6} + 2k\pi, \pi - \frac{\pi}{6} + 2k\pi$$

$$x = \frac{(2k+1)\pi}{2}, \frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi \quad (k \in \mathbb{Z})$$

(d) $(1+x)^{m+n} = (1+x)^m (1+x)^n$

On LHS coeff $x^3 = \binom{m+n}{3}$

$$\text{RHS} = \left(1 + \binom{m}{1}x + \binom{m}{2}x^2 + \binom{m}{3}x^3 + \dots \right)$$

$$\times \left(1 + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots \right)$$

$$\text{Term in } x^3 = 1 \times \binom{n}{3}x^3 + \binom{m}{1}x \times \binom{n}{2}x^2 + \binom{m}{2}x^2 \times \binom{n}{1}x$$

$$+ 1 \times \binom{m}{3}x^3$$

$$\therefore \text{Coeff } x^3 = \binom{n}{3} + \binom{m}{1} \times \binom{n}{2} + \binom{m}{2} \times \binom{n}{1} + 1 \times \binom{m}{3}$$

$$= \binom{m+n}{3}$$